

1. Count the following:

- How many different anagrams are there of the word **MISSISSIPPI**?
- How many solutions are there to $x + y + z = 12$ for x, y, z non-negative integers.
- How many strings are there of length 5 with no repeated letters.
- How many subsets of size 3 are there from $\{1, 2, \dots, 10\}$.

2. Is it possible to have a graph with the following degree sequences:

- $(3, 3, 3, 3, 3, 3, 3)$
- $(4, 2, 2, 2, 2, 2, 2)$

3. How many surjective functions are there $f : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$?

4. A *directed graph* is a pair $G = (V, E)$ where V is a set of vertices and $E \subseteq V \times V$ is a set of *ordered* pairs called directed edges. Namely, in a directed graph we can only traverse the edges in one direction. We write $u \rightarrow v$ to denote $(u, v) \in E$. A *tournament* is a directed graph $T = (V, E)$ such that for every pair of distinct vertices $u, v \in V$, exactly one of $u \rightarrow v$ or $v \rightarrow u$ is in E . A *directed Hamiltonian path* is path in this directed graph that visits every vertex exactly once. Prove that every tournament has a directed Hamiltonian path.

5. Give a combinatorial proof of $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$.